

Data Collection Plan – Measure Phase

Purpose

To collect simple, reliable data on checkout waiting times and main delay causes during the morning rush (8–10 a.m.) at the campus coffee shop.

Metric	Data Source	Who Collects	When / Frequency	Sampling Approach	Planned Sample Size
Checkout waiting time (minutes)	Manual timing sheet at register	Cashiers	Weekdays, 8–10 a.m., for 2 weeks	Every 3rd customer in line	~200 customers
Payment method (cash vs card)	Timing sheet tick box	Cashiers	Same as above	Recorded for each timed customer	Same 200 customers
Delay reason (if long wait)	Short note on timing sheet	Cashiers	Only when wait > 7 minutes	Judgment sample (all long waits)	Estimated 60–80 cases
Active cashier count	Observation	Shift supervisor	Every 15 minutes, 8–10 a.m., 2 weeks	Full capture	~80 time blocks

Data Recording Method

- Use a one-page paper timing sheet per shift.
- Each row = one observed customer.
- Columns: time joined line, time payment complete, payment method, long-wait reason (if any).
- Supervisor collects sheets at end of shift and stores them in a simple folder or shared drive.